3. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_HISTORY 10

int main() {
    char *history[MAX_HISTORY];
    int count = 0;
    char buffer[200];

    while (1) {
        printf("myshell> ");
        fgets(buffer, sizeof(buffer), stdin);

        buffer[strcspn(buffer, "\n")] = '\0';

        if (strcmp(buffer, "exit") == 0)
            break;

        if (strcmp(buffer, "history") == 0) {
            for (int i = 0; i < count; i++)
                printf("%d  %s\n", i + 1, history[i]);
            continue;
        }

        if (strlen(buffer) > 0) {
            if (count == MAX_HISTORY) {
                free(history[0]);

                for (int i = 1; i < MAX_HISTORY; i++)
                    history[i - 1] = history[i];

                count--;
            }

            history[count] = malloc(strlen(buffer) + 1);
            strcpy(history[count], buffer);
            count++;
        }
    }

    for (int i = 0; i < count; i++)
        free(history[i]);

    return 0;
}
```

```
hp@Hima:~$ nano hello.c
hp@Hima:~$ gcc hello.c -o hello
hp@Hima:~$ ./hello
myshell> ls
myshell> pwd
myshell> date
myshell> history
1  ls
2  pwd
3  date
myshell> exit
hp@Hima:~$
```